MATH 111 Calculus 1

The rules

Differentiation formulas · Derivatives of sin and cos

Week 3 · Lecture 9 · §2.3, §2.4

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. In the meantime, from the definition: the derivative of x^2 is $2x$, and of x^3 it is $3x^2$. Guess the derivative of x^{100} — and of $\sqrt{x}$.

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"Where did the Weierstrass function come from?"

From the question "what does the definition actually force?" — nothing about pictures. Today we go the other way and ask what the definition *gives*, and the answer is a short list of rules that covers almost everything you will ever differentiate.

STAND UP

"The derivative of a product is the product of the derivatives: $(fg)' = f'g'$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE, and it is the most expensive mistake in the chapter. Take $f(x) = g(x) = x$. Then $fg = x^2$, so $(fg)' = 2x$; but $f'g' = 1 \cdot 1 = 1$. Not even close. Sums behave; products do not. The true rule costs one extra term, and we shall see today exactly where that term comes from.

The easy rules — straight from the definition

Four rules (c a constant; f, g differentiable at x)

constant	**power**	**constant multiple**	**sum, difference**
$\frac{d}{dx}(c) = 0$	$\frac{d}{dx}(x^n) = nx^{n-1}$	$\frac{d}{dx}\big[cf(x)\big] = cf'(x)$	$\big[f \pm g\big]' = f' \pm g'$

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Proof of the constant rule.

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Proof of the constant multiple rule. A constant factor comes out of a limit (week 1).

$$\frac{d}{dx}\big[cf(x)\big] = \lim_{h\to 0} \frac{cf(x+h) - cf(x)}{h} = c\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = cf'(x).$$ □

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The difference $f-g$: the same proof, with minus signs. Week 1's limit laws, used at last.

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$$= n\,a^{n-1}.$$

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Before you differentiate, look

Rewrite roots and reciprocals as powers first: $\sqrt[3]{x^2} = x^{2/3}$, $\dfrac{5}{x^4} = 5x^{-4}$. Half the work in this chapter is algebra done before the rule is applied.

Question 19

What is $\dfrac{d}{dx}\left[x^3\,(x+2)\right]$?

A. $3x^2 \cdot 1 = 3x^2$

B. $4x^3 + 6x^2$

C. $3x^2(x+2)$

D. x^3

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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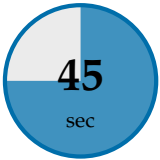
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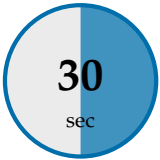
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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

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Hands down. Thank you.

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Hands up — one of you gives the reason; I repeat it aloud for everybody.

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Answer

B. Multiply out first: $x^3(x+2) = x^4 + 2x^3$, and the sum and power rules give $4x^3 + 6x^2$. A is the warm-up's mistake, $(fg)' = f'g'$. C differentiates only the first factor and D only the second — both are "half a product rule". Check any of them at $x = 1$: the true slope is 10, and A, C, D give 3, 9, 1.

The product rule

By analogy with the sum rule one is tempted to guess that the derivative of a product is the product of the derivatives — Leibniz himself guessed so. The warm-up showed that the guess is wrong ($f = g = x$). The correct formula, found by Leibniz soon after his false start, is the *product rule*.

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The product rule (the Leibniz rule)

If f and g are both differentiable, then

$$\frac{d}{dx}\big[f(x)\,g(x)\big] = f(x)\,\frac{d}{dx}\big[g(x)\big] + g(x)\,\frac{d}{dx}\big[f(x)\big], \qquad \text{i.e.} \qquad (fg)' = f\,g' + g\,f'.$$

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The proof is on the next slide: the definition of the derivative, one algebraic trick, and the limit laws.

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&= f(x)\,g'(x) + g(x)\,f'(x).
\end{aligned}
$$

□

$\lim_{h\to 0} g(x) = g(x)$, because $g(x)$ is a constant with respect to h; $\lim_{h\to 0} f(x+h) = f(x)$, because f is differentiable at x, hence continuous at x (Lecture 8).

A little history · 1675–1684

11 November 1675. In a manuscript, **Leibniz** asks our warm-up question: is $d(xy)$ the same thing as $dx\,dy$ — and $d\frac{x}{y}$ the same as $\frac{dx}{dy}$? At first it "appeared to be proved". Then, in the same manuscript: it is incorrect. "This is a difficult point."

A little history · 1675–1684

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October 1684. *Nova methodus pro maximis et minimis*, in the journal *Acta Eruditorum*: the first published work on the calculus — the rules for powers, products and quotients, *with no proofs*. Jacob Bernoulli called it an enigma rather than an explanation.

The quotient rule

Quotient rule

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You may stop there — expanding makes it no clearer.

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Proof. Let $F(x) = \dfrac{f(x)}{g(x)}$, with $g(x) \neq 0$; g is continuous at x (Lecture 8), so $g(x+h) \neq 0$ for small h.

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continued →

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Again g is continuous by Lecture 8, so $\lim_{h\to 0} g(x+h) = g(x)$; the limit of the bottom is $[g(x)]^2 \neq 0$, so the limit law for a quotient applies. $\lim_{h\to 0} g(x) = g(x)$ and $\lim_{h\to 0} f(x) = f(x)$: constants with respect to h.

STAND UP

"To differentiate a quotient you must use the quotient rule."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE. You must use it when nothing simpler works — but look first. For $x \neq 0$, $\dfrac{x^4 - 5x^2}{x^2} = x^2 - 5$, so the derivative is $2x$ for every $x \neq 0$, in one line; the quotient rule would take five and give the same answer. At $x = 0$ there is no derivative: the function is not defined there — its graph is the parabola $y = x^2 - 5$ with a hole at $(0, -5)$.

Simplifying changes the formula, not the domain. Likewise $\dfrac{7}{x^3} = 7x^{-3}$, again for $x \neq 0$.

The rule is a tool, not a duty.

The list so far

Everything from §2.3

$$\frac{d}{dx}(c) = 0 \qquad \frac{d}{dx}(x^n) = nx^{n-1} \quad \text{(every real } n\text{)}$$

$$(cf)' = cf' \qquad (f \pm g)' = f' \pm g'$$

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so $y' = \frac{3}{2}x^{1/2} + x^{-1/2} = \dfrac{3\sqrt{x}}{2} + \dfrac{1}{\sqrt{x}}$ — no product rule needed at all.

Question 20

Suppose $f(3) = 2$, $f'(3) = -4$, $g(3) = 5$, $g'(3) = 6$. What is $(fg)'(3)$?

A. -24

B. -8

C. 32

D. Not enough information.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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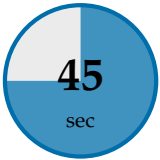
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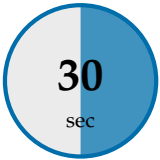
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Hands down. Thank you.

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Question 20

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Hands up — one of you gives the reason; I repeat it aloud for everybody.

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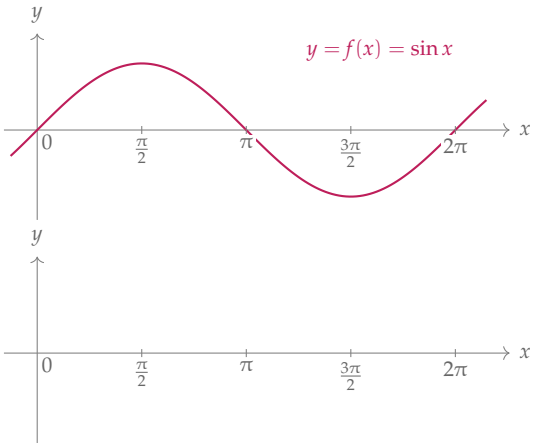
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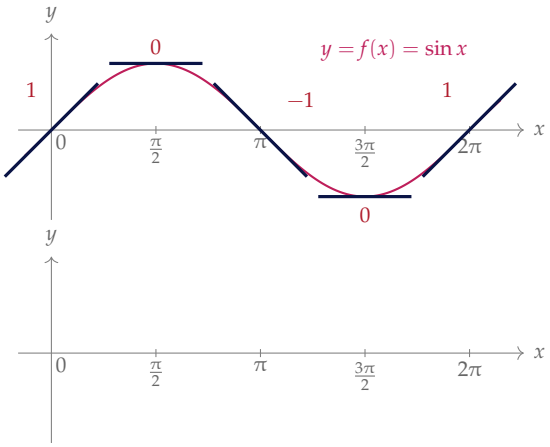
Answer

B. $(fg)'(3) = f(3)g'(3) + g(3)f'(3) = 2 \cdot 6 + 5 \cdot (-4) = 12 - 20 = -8$. A is $f'(3)g'(3)$, the warm-up's error. C is $12 + 20$ — the product rule with the minus sign of $f'(3)$ thrown away. D is tempting because we have no formulas for f and g; the rule needs only these four numbers, which is exactly what makes it a rule.

Derivatives of trigonometric functions

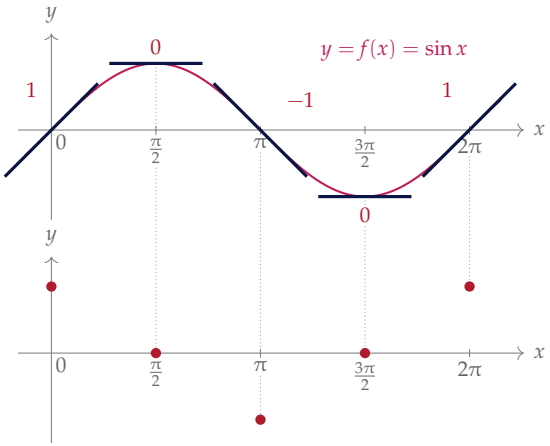


The graph of $\sin x$ and of its derivative. To sketch f', read the slope of the tangent to the sine curve — the method of Lecture 8.



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Slopes at $0,\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2},\ 2\pi$:
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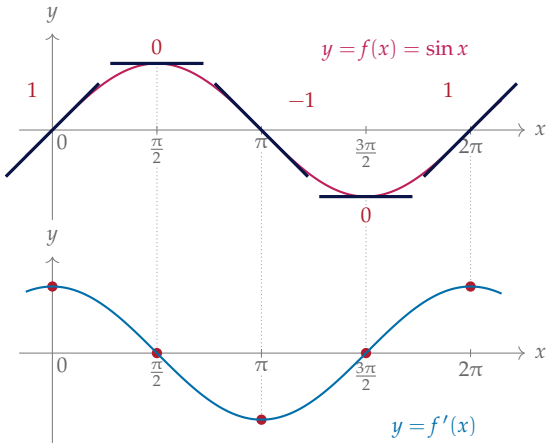


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Join the points: the graph of f' looks like the cosine curve.

A guess from a picture — not yet a proof. The proof is next.

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Two of the four limits are easy: x is fixed while $h \to 0$, so $\lim_{h\to 0} \sin x = \sin x$ and $\lim_{h\to 0} \cos x = \cos x$. The other two are the special limits (h in radians; their proofs are in §2.4 — homework): $\lim_{h\to 0} \frac{\sin h}{h} = 1$, $\lim_{h\to 0} \frac{\cos h - 1}{h} = 0$.

Into (1): $f'(x) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x$. □

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The *word* radian is younger: it is in print on 5 June 1873, in examination questions set by James Thomson, Lord Kelvin's brother, at Queen's College, Belfast.

The whole trigonometric table

Six derivatives

$(\sin x)' = \cos x$ $\quad$ $(\tan x)' = \sec^2 x$ $\quad$ $(\sec x)' = \sec x \tan x$

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Homework exercise: prove the formulas for the rest of the trigonometric functions — cos in the same way as sin, with $\cos(x+h) = \cos x \cos h - \sin x \sin h$; cot, sec, csc by the quotient rule.

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This week:

Homework: §2.3 and §2.4 — read and work out the examples and proofs. Next: the chain rule.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"$(fg)'$ is not $f'g'$ because . . ." (one sentence, or one example).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.